

Project Status and Provenance Record: AlphaSAXS (Phase 1)

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Subject: Documentation of Phase 1 Milestones, Intellectual Property Lineage, and Technical Deliverables

Executive Summary

As of June 2026, Phase 1 of the AlphaSAXS computational development is concluding. This document serves as the formal record of the project's data provenance, structural development timeline, and the specific intellectual and technical contributions of the Phase 1 core technical developers (Feng Yu and Stephanie Prince) under the direction of the senior PIs. This timeline establishes the finalized baseline methodologies and datasets that will serve as the foundation for future work on the system.

Phase 1 Development Timeline & Technical Contributions

Late 2023: Project Inception & Team Formation

- **Oct - Dec 2023:** Stephanie Prince and Feng Yu officially join the AlphaSAXS initiative as the core technical developers handling computational architecture and model implementation.

Q1 2024: Dataset Standardization & Architecture Design

- **Jan 2024:** Established the PDB70 database as the primary training data and curated 40 apo-holo pairs for the test dataset. Identified and diagnosed the "missing loop" problem within the PDB training dataset. (*Primary Developer: Feng*)
- **Mar 2024:** Developed and deployed the v1 loop-fixing script on the HPC. (*Primary Developer: Feng*) Established AlphaFlow and OpenFold as the primary architectures for introducing Multiple Sequence Alignment (MSA) and pair-representation cross-attention units. (*Primary Developer: Stephanie*)

Q2 - Q4 2024: Data Generation, Compute Allocation & Initial Training

- **May 2024:** Generated the MSA dataset for training and created the normal mode analysis for PDB70. (*Primary Developer: Stephanie*) Successfully finalized "PDB70

Fixed Version B." Secured critical NERSC GenAI computational allocation under Feng Yu's name. (*Primary Developer: Feng*)

- **Nov 2024:** Successfully trained and delivered the first complete iteration of the AlphaSAXS model. (*Primary Developers: Stephanie, Feng*)

2025: Optimization, Open-Source Licensing & Dissemination

- **Feb 2025:** AlphaSAXS research submitted to the GEM Workshop at the 2025 ICLR conference (Accepted and presented in April 2025). (*Authors/Contributors: Feng, Stephanie*)
- **Mar 2025:** Formulated, improved, and computationally integrated the differentiable SAXS loss function into the model. (*Primary Developer: Stephanie*)
- **Oct 2025:** Developed and introduced the ensemble generation algorithm to improve structural prediction accuracy. (*Primary Developer: Feng*) Successfully secured an open-source license from LBNL for the Beta version of the AlphaSAXS model. (*Primary Developer: Stephanie*)
- **Dec 2025:** Stephanie Prince transitions off the team, concluding her active technical contributions to the project.

2026: Publication, Future Proposals & Phase 1 Conclusion

- **Apr 2026:** Finalized manuscript detailing the AlphaSAXS Phase 1 architecture and results submitted. The paper is currently under peer review for *Nature Communications*. (*Authors/Contributors: Feng, Stephanie*)
- **May 2026:** Drafted and submitted the proposal for the "Genesis" mission, establishing the theoretical and computational framework for a multi-modal system integrating AlphaSAXS with neutron scattering. (*Primary Contributor: Feng*)
- **Jun 2026:** Feng Yu officially transitions off the AlphaSAXS project, marking the end of Phase 1 technical development.

Current Status & Repository Deliverables

The datasets, computational scripts, and model architectures associated with Phase 1 are formally concluded.

This repository serves as the static baseline for any forthcoming development on Phase 2 (Multimodal Structure Refinement System).